$$\mathbf{I} = \int dp|p\rangle\langle p|. \qquad (8.22)$$

Let's suppose that we know the wave function of the abstract vector |Ψ> in the position representation. By definition, it is equal to

$$\psi(x) = \langle x|\Psi\rangle. \qquad (8.23)$$

Now suppose we want to know the wave function $\tilde{\psi}(p)$ in the momentum representation. Here are the steps laid out in detail:

- First, use the definition of the momentum-representation wave function:

$$\tilde{\psi}(p) = \langle p|\Psi\rangle.$$

- Now, insert the unit operator between the bra- and ket-vectors, in the form given in Eq. 8.21:

$$\tilde{\psi}(p) = \int dx\langle p|x\rangle\langle x|\Psi\rangle.$$

- The expression <x|Ψ> is just the wave function $\psi(x)$, and <p|x> is given to us by the second equation of Eqs. 8.18:

$$\langle p|x\rangle = \frac{1}{\sqrt{2\pi}} e^{\frac{-ipx}{\hbar}}.$$

- Putting it all together, we find that